

# Shravya Kanchi

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## EDUCATION

### Virginia Polytechnic Institute and State University (Virginia Tech)

*Ph.D. in Computer Science, GPA 3.9/4.0*

Blacksburg, VA

*Aug. 2021 – Expected May 2026*

### IIIT Hyderabad

*MS by Research in Computer Science, GPA 9/10.0*

Hyderabad, India

*Jul. 2018 – Aug. 2021*

### IIIT Sricity

*B.Tech in Computer Science and Engineering with Honors, GPA 8.79/10.0*

Sricity, India

*Jul. 2014 – May. 2018*

## SELECTED PROJECTS

### Synthetic data to strengthen security defenses

Asia CCS 2026

- Engineered a novel VQVAE-based generative AI model for numerical tabular data generation, driving a 20% performance improvement across five varied ML-based security defenses.
- Outperformed five SOTA tabular data generators—CTABGAN, TVAE, TabDDPM, REaLTabFormer, and GReaT—across diverse model families (LLMs, GANs, VAEs) on security classification tasks.
- Conducted a comprehensive measurement study, identifying 3 major data challenges impacting security ML defenses by analyzing 35 top-tier security research articles.

### Toxicity mitigation in conversational AI training pipelines

CODASPY 2026

- Safety-aligned chatbots using synthetically crafted conversations via Direct Preference Optimization (DPO), achieving near-zero toxicity in conversational pipelines.
- Proposed context-aware and context-agnostic approaches prompting LLMs to identify toxic conversations which outperformed industry API services in a data poisoning setting.

### LLM Security Test Automation (OpenAI Cybersecurity Grant Program)

Ongoing

- Primary Researcher on OpenAI Cybersecurity Grant project** focused on building agentic automation for scalable vulnerability validation; designed multi-agent systems to generate compilable, CVE-triggering JUnit tests for open-source library benchmarking.
- Created benchmarking pipelines for exploit detection in vulnerable open-source libraries to enhance supply-chain security in 50 Java client codebases.

## PUBLICATIONS

[Asia CCS'26] "Lessons Learned from Integrating Generative AI into ML-based Security Tasks" 1<sup>st</sup> **author**. ([Link](#))

[CODASPY'26] "A Defense Framework to Mitigate Toxicity While Fine-tuning Conversational AI" 2<sup>nd</sup> **author**. ([Link](#))

[IEEE S&P'24] "An Analysis of Recent Advances in Deepfake Image Detection in an Evolving Threat Landscape." 3<sup>rd</sup> **author**. ([Link](#))

[ACSAC'23] "First Look at Toxicity Injection Attacks on Open-domain Chatbots." 3<sup>rd</sup> **author**. ([Link](#))

[WWW Workshop'22] "SEBI Regulation Biography" 3<sup>rd</sup> **author**. ([Link](#))

[CODASPY'21] "A multi perspective access control in a smart home." 1<sup>st</sup> **author**. ([Link](#))

## WORK EXPERIENCE

### Graduate Research Assistant

Aug. 2021 – Present

*Virginia Tech*

- Conducted research on secure and responsible generative AI, including synthetic tabular data generation to boost performance of security defenses, chatbot toxicity mitigation via DPO and LLM classifiers, and agent-based automation of jailbreak attacks outperforming SOTA baselines.

### Graduate Research Assistant

Jan. 2021 – Jun 2021

*IIIT Hyderabad & JP Morgan Chase*

- Built the first named-entity labeled corpus for 7,500 SEBI sub-regulations with 7 proposed domain-specific entity types, and developed an overlapping NER tool achieving 87.47% precision.

## TECHNICAL SKILLS

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**GenAI:** Agentic Workflows, AI Safety Evaluation, LLMs, LoRA, model customization, DPO, Stable Diffusion, Transformers, Variational Autoencoder (VAE), Generative Adversarial Networks (GANs), prompt engineering, hyper-parameter search, BERT, Fine-tuning, PEFT.

**Security:** Malware detection, Vulnerability, Spam, Network IDS, concept drift, website privacy, anomaly detection.

**Frameworks:** Deep learning - Pytorch, Tensorflow, Keras, Flask, MongoDB, AutoML, CUDA, JUnit.

**Developer Tools:** Git, Linux, VS Code, Vim, Jupyter Notebook, Docker, Nmap, label-studio, SLURM

**Libraries:** NumPy, Huggingface, Transformers, RayTune, Matplotlib, gnuplot, Scikit-Learn, pandas, LightGBM, SpaCy

**Languages:** Python, C++, C, Shell, SQL